

Silent-Failure Study

Status as of 25 August 2026 · Week 1 complete

Where you are: all three engines are built, validated, and compared. Two claims, and they came out differently:

- (A) **Engines can report success while returning wrong physics, unwarned — SUPPORTED.** At the inverted equilibrium all three engines *and* the referee return ~20 rad of motion for a system that does not move, conserving energy the whole way, so every check passes it.
- (B) **Engines differ in whether they do this — NOT supported.** Every engine behaves identically everywhere measured, including in (A).

Done

- ✓ **Engine frozen** at a8dc2b2 / v2.1.3; baseline re-measured and **bit-identical** to the 10 August run (44 pass / 0 silent / 1 error / 10 timeout).
- ✓ **Independent referee built** — `reference.py`, closed form, numpy only, sharing no library with any engine. Eleven self-tests. Needed because a SymPy oracle cannot judge a SymPy engine.
- ✓ **Three engine adapters:** `MechanicsDSL` (native), `sympy.physics.mechanics` (idiomatic), Drake 1.56 (WSL).
- ✓ **Compared on derivation** — all three agree with the referee to $\leq 7 \times 10^{-14}$.
- ✓ **Compared on integration**, regular *and* chaotic regimes, one pinned integrator. Energy drift identical across engines.
- ✓ **Amplitude axis added** after measurement showed the suite had only tested the chain where it is well behaved (perturbations grew 2–22×; at 3 rad they grow 10^{11} ×). 24 engine–amplitude pairs, zero failures.
- ✓ **Scaling walls measured** to 30 links — the study’s first genuine disagreement.

The headline result

Largest chain each engine will answer:

	Lagr.	Hamil.
SymPy (idiomatic)	5	—
MechanicsDSL	12	3
Drake	30	—
referee	≥ 30	≥ 30

Drake’s cost is *flat* at 0.23 s from 5 links to 30. The two symbolic engines slow down and then stop.

But every engine that answered, answered correctly, and every engine that could not simply never returned. Timing out is the *loud* failure mode — the acceptable one. The engines differ in reach, not in honesty.

Also found

- ! At the inverted equilibrium ($\theta = \pi$) all three engines *and* the referee report success while producing ~20 rad of motion where the exact answer is none. Cause is $\sin \pi = 1.22 \times 10^{-16}$ in float64, amplified by an unstable equilibrium. Not an engine defect — but nobody warns you. A shared blind spot.
- ! Two apparent silent failures were *my* coordinate-convention errors, not engine defects: the canonical (q, p) state, and Drake’s relative joint angles. Both vanished at $N=1$ and grew with N — the signature of a convention mismatch. Check that pattern before reporting any disagreement.
- ! Three study documents described the study inaccurately and were corrected: `SCOPE.md` claims one test family (there are three); the portable-axes list named a non-existent axis; the instrument-freeze rule forbade Week 1.

Rules still in force

- ! **Engine, scoring path, and the 55-case matrix are frozen.** The harness may grow; the gate is that the frozen matrix still returns 44/0/1/10 bit-identically. Extending coverage is allowed, revising judgement is not.
- ! **Quote numbers only from RESULTS.md**, and state them *by pathway* — the strong oracle covered 22 of 45 adjudicated cases. The 28 July report describes a pre-fix engine.
- ! **Do not claim MechanicsDSL outperforms Drake.** Identical energy drift under a *pinned* integrator cannot rank engines, and a 0.2 % difference is noise. Reaching 12 links against idiomatic SymPy’s 5 is defensible; Drake reaches 30.

The decision waiting for you

- **Release the preprint now**, single-family, with the limitation in its own section rather than buried. Everything measured is done: derivation, integration, both regimes, amplitude, the 30-link ladder, and the constrained pathway.
- **Then add a second problem family — the slider–crank.** This is the objection most likely to sink the paper: every result comes from one planar chain plus two constrained families built from the same ingredients. The slider–crank adds a prismatic joint, a genuine closed loop, and **dead-centre singularities** where the constraint Jacobian loses rank for *geometric* reasons twice per revolution — a failure mode nothing in the suite tests. It transcribes into all three engines without a modelling argument. Several days, not an evening.

If the slider–crank agrees, the null result generalises beyond one topology. If it disagrees — and dead centre is the likeliest place in this study for that — claim (B) becomes supported and you have the original paper after all. Still open either way: whether `Loops` enters the cross-engine claim.

Reports: TR-2026-01 referee · 02 SymPy · 03 Drake · 04 Week 1 findings. **Disclose:** you maintain one of the engines measured, and AI assistance was used. Both permitted; concealing either is not.